

E0 — Frozen model baseline
Question: Can Gemma beat random?
Result: No; click actions often do nothing

Direct model decisions are insufficient

E1 — Harness memory
Question: Does remembering failures help?
Result: Pruning improves exploration, but solving only matches random

Memory helps search but not understanding

E2 — Control-effect descriptions
Question: Does explaining button effects help?
Result: No; dead moves increase

Providing information does not ensure useful reasoning

E3 — LEARN vs WIN actions
Question: Can the model
separate
learning from goal-directed
actions?

Result: Stopped at tune;
dead moves increase and
labels are ignored

Prompt labels and a diary do not
create epistemic control

**E4 — Random +
dead-action pruning**

Question: How far can a
simple policy go?

Result: Public clears double,
but hidden score does not
improve

Simple search is strong locally but
does not generalize

Capability gap after E4
Simple filtering cannot
reliably infer
mechanics, goals, causality,
or plans

Test whether slow reasoning
supplies the missing capabilities

Test whether better fast action
selection is sufficient

